MATH9102

LOGISTIC REGRESSION

Sources used in creation of this lecture:
Statistics and Data Analysis, Peck, Olsen and Devore; Discovering Statistics Using R, Field, Miles and Field; Understanding Basic Statistics, Brase and Brase;SPSS Survival Manual, Julie Pallant

What is logistic regression?

A method used when the outcome is categorical, usually binary

Used to estimate the probability of belonging to a class

Works with categorical or continuous predictors

Output is always between 0 and 1

What does logistic regression try to do?

Model the probability of an event occurring depending on the values of the independent variables

Estimate the probability that an event occurs vs the probability that the event does not occur

Predict the effect of a series of variables on a binary response variable

Classify observations by estimating the probability that an observation is in a particular category or not.

Starting Equation

In logistic regression, we first build an ordinary **linear equation**:

- $t = b_0 + b_1x_1 + b_2x_2 + \cdots$

This looks just like a linear regression

- **t can take any value from –∞ to +∞**

Logistic regression does **not** model probability directly

It models **t**, which is later transformed into a probability

This is the input to the logistic function.

Log Odds

$t = \log\left(\frac{p}{1-p}\right)$

- This is the **logit** function
- It transforms probability (0–1) into **log-odds** (−∞ → +∞)

If t = 0

- →odds = 1 → p = 0.5
- The model is **uncertain** (50/50 chance).

If t > 0

- The **odds** of the event increase
  → p > 0.5
- The model leans toward **“Yes, the event happens.”**

If t < 0

- The **odds** of the event decrease
  → p < 0.5
- The model leans toward **“No, the event does not happen.”**

From Log Odds to Odds

To remove the log, we exponentiate:

- $\text{odds} = e^t$

Meaning of odds:

- $\text{odds} = \frac{p}{1-p}$
- odds = 1 → 50/50
- odds > 1 → event more likely
- odds < 1 → event less likely

From Odds to Probability

Content:

$$p = \frac{odds}{1+odds}$$

Substituting odds = e^t:

$$p = \frac{e^t}{1+e^t}$$

This produces a value between **0 and 1**.

This is how logistic regression turns any linear equation into a valid probability.

Linear equation (t)

↓

Log-odds = logit(p)

↓

Odds = e^t

↓

Probability = odds / (1 + odds)

Concept	**Range**	**Meaning**
Probability	0 → 1	Chance something happens
Odds	0 → ∞	Ratio of happening vs not happening
Log-odds	-∞ → ∞	Linear scale used by model

Simple Example

Binary Logistic Regression Example Predicting Exam Success from Study Hours

Research question:

Does the number of hours a student studies predict whether they pass an exam?

Outcome (dependent variable):

- **pass_exam**
  - 1 = pass
  - 0 = fail

Predictor (independent variable):

- **study_hours** (continuous)

This is a classic situation for logistic regression because the outcome is binary.

Example – what logistic regression does?

Logistic regression estimates how the probability of passing changes with each extra hour studied:

$$\log\left(\frac{p}{1-p}\right) = b_0 + b_1(study\_hours)$$

Where:

b_0 :baseline log-odds of passing

b_1 :change in log-odds for each additional hour of study

Example – Interpreting the coefficient

If the model estimates $b_1 = 0.25$

This means:

- For each extra hour studied, the **log-odds** of passing increase by 0.25
- We can get the odds ratio by exponentiating the log-odds
- The **odds ratio** = $e^{0.25} = 1.28$

Interpretation:

- Each additional hour of study increases the odds of passing by **28%**.

logit

Logistic regression predicts probabilities, which must stay between: $0 \leq p \leq 1$

But a linear equation can take *any* value from: $-\infty \; to \; +\infty$

So we transform probability into something a linear model can handle.
That transformation is the **logit**, which converts probabilities into **log-odds**.

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

Where:

- p= probability of the event happening
- $\frac{p}{1-p}$ =**odds** of the event
- logit(p) = **log-odds**

Example - Predicting Exam Success from Study Hours and Attendance

Research question:

Do study hours and class attendance together predict whether a student passes an exam?

Outcome (binary):

- **pass_exam**
  - 1 = pass
  - 0 = fail

Predictors:

- **study_hours** (continuous)
- **attendance** (categorical)
  - 0 = low attendance (<70%)
  - 1 = high attendance (≥70%)

- one continuous predictor
- one categorical predictor

Example- Logistic Regression Model

The model becomes:

$$\log\left(\frac{p}{1-p}\right) = b_0 + b_1(study\_hours) + b_2(attendance)$$

Interpretation:

b_1 :effect of each additional hour studied

b_2 :effect of being in the high-attendance group (compared to low attendance)

Predictor	Coefficient	Interpretation
(b_0)	–2.0	baseline log-odds of passing when hours = 0 and attendance = low
(b_1)	0.20	each hour studied increases log-odds by 0.20
(b_2)	1.10	high attendance increases log-odds by 1.10

Suppose we get this output.

Convert to odds ratios:

OR for study_hours = $e^{0.20} = 1.22$
→each extra hour increases odds of passing by **22%**

OR for attendance = $e^{1.10} = 3.00$
→high-attendance students have **3 times the odds** of passing compared to low-attendance students

Example – Predicted Probabilities

Let's compare predicted probabilities for different scenarios.

Low attendance (0), 5 hours of study:

- $\text{"logit"} = -2 + 0.2(5) = -1$
- p=0.27
- **27% chance of passing**

High attendance (1), 5 hours of study:

- $\text{"logit"} = -2 + 0.2(5) + 1.1 = 0.1$
- p=0.525
- **52% chance of passing**

High attendance, 10 hours of study:

- $\text{logit} = -2 + 0.2(10) + 1.1 = 1.1$
- p=0.75
- **75% chance of passing**

Logistic regression

Odds ratios are used to calculate probabilities in a logistic regression.

Predictor variables can be either continuous or categorical.

Logistic regression does *not* require the predictors to have a linear relationship with the *outcome*.

- BUT it *does* require each predictor to be linearly related to the **logit of the probability**.

It is sensitive to high correlation between predictor variables (multicollinearity).

Full Worked Example

YOUTHCOHORT DATASET PAUL CONNOLLY

MATH9102-W11-LOGREG.QMD

Plus accompanying html output

Example - Youthcohort Dataset

Taken from Quantitative Data Analysis in Education, Paul Connolly

Dataset: youthcohort.sav

Research question:

- What factors predict the likelihood that a student will answer yes when asked if they sat French at GCSE?
- Note:
  - We will not build a model to fully explore this question – we will use only selected predictors to illustrate how logistic regression can be interpreted

Example

This model explores whether parental educational achievement helps predict a young person’s likelihood of answering yes to whether they sat French at GCSE level.

- satfren is the binary outcome
  - Yes = sat GCSE French
  - No = did not
  - Converted to a factor satfren_num 1 = sat GCSE French, 0 = No did not
- s1pared is the sole predictor
  - At least one parent with degree
  - At least one parent with A-level
  - Neither parent with A-level
  - This will allow us to estimate how differences in parental education are associated with the probability that a student sat GCSE French.
  - This has been relevelled to make Neither Parent with an A-level as the reference category

Dummy Coding

Dummy coding is a method used to include **categorical variables** in statistical models

- It converts a categorical variable with *k* categories into **k–1 binary (0/1) variables**.

Why k–1?

- Because one category is chosen as a **reference category**, and the others represent comparisons against that reference.

Any categorical predictors included in the logistic regression model must be dummy coded

- R will automatically do this, but if you want to choose your reference category you need to recode yourself.

Dummy Coding - Example

Education	Undergrad (D1)	Postgrad (D2)
High school	0	0
Undergraduate	1	0
Postgraduate	0	1

Suppose you have a categorical predictor **Education level:**

- High school
- Undergraduate
- Postgraduate

Choose **High school** as the **reference** category.

The logistic regression model becomes:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 D1 + \beta_2 D2$$

Interpretation:

$\boldsymbol{\beta_0}$ = log-odds for the **reference group** (High school)

$\boldsymbol{\beta_1}$ = difference in log-odds between **Undergrad** and High school

$\boldsymbol{\beta_2}$ = difference in log-odds between **Postgrad** and High school

Exponentiating coefficients gives odds ratios.

Dummy Coding - Example

Education	Undergrad (D1)	Postgrad (D2)
High school	0	0
Undergraduate	1	0
Postgraduate	0	1

The logistic regression model becomes:

- $\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 D1 + \beta_2 D2$

Interpretation:

β_0 = log-odds for the **reference group** (High school)

β_1 = difference in log-odds between **Undergrad** and High school

β_2 = difference in log-odds between **Postgrad** and High school

Say:

$\beta_1 = 0.80 \rightarrow \exp(0.80) \approx$ **2.22**

This means:
Participants with an undergraduate degree have **2.22 times higher odds** of the outcome **compared to the high school group**, holding other variables constant.

Step 1: Build the Model

```
logmodel1 <- glm(satfren_num ~
s1pared, data =youthdf, na.action =
na.exclude, family =
binomial(link=logit))
```

- binomial(link=logit) specifies the type of regression model being used within the glm (generalized linear model) function
- family = binomial(link=logit): This argument specifies that the model is a logistic regression model.
  - The binomial family is used when the dependent variable (in this case, satFrench) is binary (0 or 1, yes or no, etc.).
  - binomial: Indicates that the dependent variable follows a binomial distribution, which is common for binary outcome variables.
  - link=logit: Specifies the link function as logit, which is the standard choice for logistic regression.
  - This function models the log odds of the dependent variable.

Step 2: Generate Statistics to Assess Model Fit and usefulness (a) Summary

```
summary(logmodel1)
```

An omnibus test of model is used to check that the new model (with explanatory variables included) is an improvement over the baseline model (no predictors).

It uses **chi-square** tests to see if there is a significant difference between the baseline model (null) and the model created

Step 2: Generate Statistics to Assess Model Fit and usefulness (b) Omnibus test

Likelihood Ratio (LR) Test

- Does the fitted model significantly improve fit over the null model?

Use lrtest from lmtest package

The Chi-square statistic is 166.19 df=2, N=13201, and the p value is is statistically

```
#Chi-square plus significance
lmtest::lrtest(logmodel1)
```

```
Likelihood ratio test

Model 1: satfren_num ~ s1pared
Model 2: satfren_num ~ 1
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   3 -8978.9
2   1 -9062.0 -2 166.19  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Step 2: Generate Statistics to Assess Model Fit and usefulness (c) Psuedo R2

In linear regression:

- R^2 = proportion of variance in the outcome explained by the model
- Outcome is continuous → variance is well-defined
- A linear R^2 of 0.30 = 30% of variance explained

In logistic regression:

- Outcome is 0/1
- Variance is fixed by the binomial distribution
- So "variance explained" is not meaningful

  ordinary R^2 cannot be computed
- Logistic regression uses **likelihoods** instead of sums of squares.
- **Pseudo R^2** measures are built from:
  - The likelihood of the null model (no predictors)
  - The likelihood of the fitted model
  - They tell us:
  - “How much better does the model fit the data compared to a baseline model?”
- A logistic pseudo R^2 of 0.30 = **not** 30% of variance explained
  - Instead, it indicates that **the model fits substantially better than the null model**.

.

Step 2: Generate Statistics to Assess Model Fit and usefulness (c) Psuedo R2

```
#Pseudo R Squared
DescTools::PseudoR2(logmodel1, which="CoxSnell")
```

```
  CoxSnell
0.01250994
```

```
DescTools::PseudoR2(logmodel1, which="Nagelkerke")
```

```
Nagelkerke
0.01675513
```

```
performance::r2(logmodel1) # Will return Tujur's Pseudo R2
```

```
# R2 for Logistic Regression
  Tjur's R2: 0.013
```

Cox and Snell R^2 Nagelkerke R^2 and Tijur R^2 are pseudo R^2 statistics

They are an effect size measurement – in this case between 1.3% and 1.7%

They tell us how much this model fit has improved in comparison to the baseline model (one with no predictors).

Model fit: how well the model represents the data

Step 3: Calculate Odds and Predicted Probabilities

The **Estimate** column from the output in Step 2a shows the coefficients in **log-odds** form.

Students whose parent(s) have *at least one A-level* have log-odds of sitting French that are **0.25 higher** than students whose parents **have no A-level** (the reference group).

Students whose parent(s) have a *degree* have log-odds of sitting French that are **0.53 higher** than students whose parents **have no A-level**.

```
Call:
glm(formula = satfren_num ~ s1pared, family = binomial(link = logit),
    data = youthdf, na.action = na.exclude)

Coefficients:
                                      Estimate Std. Error z value Pr(>|z|)
(Intercept)                            0.02544    0.02491   1.021    0.307
s1paredAt least one parent with A-level 0.24716    0.04545   5.439 5.37e-08
s1paredAt least one parent with degree  0.52779    0.04144  12.736  < 2e-16

(Intercept)
s1paredAt least one parent with A-level ***
s1paredAt least one parent with degree  ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 18124  on 13200  degrees of freedom
Residual deviance: 17958  on 13198  degrees of freedom
AIC: 17964
```

Getting the odds

We have the log odds at this point, but we need to calculate the actual odds to be able to calculate the odds ratios.

Odds Ratio: Indicates the change in odds resulting from a unit change in the predictor.

- OR > 1: Predictor ↑, Probability of outcome occurring ↑.
- OR < 1: Predictor ↑, Probability of outcome occurring ↓.

For continuous predictors, a “unit change” means +1 in the predictor.

For **categorical predictors**, the “unit change” means:

- **Moving from the reference category (0)**
- **to the category that dummy variable represents (1)**
- Each dummy variable compares:
  - Odds *of outcome in category j vs.Odds of outcome in reference category*

Odds Ratio

Continuous Predictor:

$$\frac{\text{Odds}(\textit{after 1 unit change in the predictor})}{\textit{Original Odds}}$$

Categorical Predictor:

$$\frac{\text{Odds(category } \textit{of interest})}{\textit{Odds(reference category)}}$$

Getting the odds

When interpreting the **odds ratios**, there are two things to look for:

Is the odds ratio greater than 1 or less than 1?

- OR > 1 → higher odds of sitting French compared to the reference group
- OR < 1 → lower odds compared to the reference group

```r
# Odds ratios only (clean output)
parameters::parameters(logmodel1, exponentiate = TRUE) %>%
  as.data.frame() %>%
  dplyr::select(Parameter, OddsRatio = Coefficient)
```

```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.

                                Parameter OddsRatio
1                             (Intercept)  1.025762
2 s1paredAt least one parent with A-level  1.280386
3  s1paredAt least one parent with degree  1.695187
```

Getting the odds

Students with at least one parent who has an **A-level** have **1.28 times the odds** of sitting French **compared to students whose parents have no A-level.**

In percentage terms:

1.28 = 28% higher odds

So we can say the odds of sitting French are **28% higher** for a student with at least one parent with an A level compared to the reference category.

```r
# Odds ratios only (clean output)
parameters::parameters(logmodel1, exponentiate = TRUE) %>%
  as.data.frame() %>%
  dplyr::select(Parameter, OddsRatio = Coefficient)
```

```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.

                                Parameter OddsRatio
1                             (Intercept)  1.025762
2 s1paredAt least one parent with A-level  1.280386
3  s1paredAt least one parent with degree  1.695187
```

Getting the odds

Students with at least one parent who has a **degree** have **1.70 times the odds** of sitting French **compared to students whose parents have no A-level.**

In percentage terms:

1.70 = 70% higher odds

The odds of sitting French are **70% higher** for a student with at least one parent with a degree than the reference category.

```
# Odds ratios only (clean output)
parameters::parameters(logmodel1, exponentiate = TRUE) %>%
  as.data.frame() %>%
  dplyr::select(Parameter, OddsRatio = Coefficient)
```

```
Profiled confidence intervals may take longer time to compute.
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                                Parameter OddsRatio
1                             (Intercept)  1.025762
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```

Getting the odds

Intercept

You usually don’t interpret the intercept in logistic regression because it represents:

The odds of sitting French for a student in the **reference category** with all predictors equal to zero

Since the predictor here is categorical, the intercept is not meaningful on its own and can be ignored for interpretation.

```
# Odds ratios only (clean output)
parameters::parameters(logmodel1, exponentiate = TRUE) %>%
  as.data.frame() %>%
  dplyr::select(Parameter, OddsRatio = Coefficient)
```

```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.

                                Parameter OddsRatio
1                             (Intercept)  1.025762
2 s1paredAt least one parent with A-level  1.280386
3  s1paredAt least one parent with degree  1.695187
```

Interpreting the output

Reporting odds ratios for categorical predictors we must remember that we are comparing the odds for two categories.

For more than two we are comparing each category to the reference category (value 0) – which we specify in the regression.

For odds ratios less than 1 we can invert these and report the inversion to help interpretation.

Getting the Odds

Is the effect statistically significant?

- Look at the p-value for each coefficient
- $p < .05 \rightarrow$ statistically significant

The **z statistic or Wald statistic** is calculated to get the significance of the predictor.

What you get from this column is whether the effect of the predictors is positive or negative.

```
Call:
glm(formula = satfren_num ~ s1pared, family = binomial(link = logit),
    data = youthdf, na.action = na.exclude)

Coefficients:
                                     Estimate Std. Error z value Pr(>|z|)
(Intercept)                           0.02544    0.02491   1.021    0.307
s1paredAt least one parent with A-level  0.24716    0.04545   5.439 5.37e-08
s1paredAt least one parent with degree   0.52779    0.04144  12.736  < 2e-16

(Intercept)
s1paredAt least one parent with A-level ***
s1paredAt least one parent with degree  ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

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AIC: 17964

Number of Fisher Scoring iterations: 4
```

Getting the Odds

The **Pr(>|z|)** column shows the two-tailed p-values testing the null hypothesis that the coefficient is equal to zero (i.e. no significant effect).

The usual cut-off value is 0.05.

From this model it appears that parental education does have a significant effect on the log-odds ratio of the dependent variable.

```
Call:
glm(formula = satfren_num ~ s1pared, family = binomial(link = logit),
    data = youthdf, na.action = na.exclude)

Coefficients:
                                        Estimate Std. Error z value Pr(>|z|)
(Intercept)                              0.02544    0.02491   1.021    0.307
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Number of Fisher Scoring iterations: 4
```

Getting the predicted probability

Neither parent has an A-level → Probability = 0.506

This is the baseline probability.

Confidence interval: **49.4% to 51.9%**

Students whose parents have **no A-levels** have a **50.6% chance** of saying they sat French.

$$P(Y) = \frac{1}{1+e^{-(b0+b1X1+b2X2)}}$$

We have the coefficients so we can calculate the predicted probability

We can use a function from the emmeans package (emmeans) to get the predicted probabilities for each predictor:

```
# Predicted probabilities by s1pared
emmeans::emmeans(logmodel1, ~ s1pared, type = "response")
```

```
s1pared                              prob      SE  df asymp.LCL asymp.UCL
Neither parent with A-level         0.506 0.00623 Inf     0.494     0.519
At least one parent with A-level    0.568 0.00933 Inf     0.549     0.586
At least one parent with degree     0.635 0.00768 Inf     0.620     0.650
```

Getting the predicted probability

At least one parent has an A-level → Probability = 0.568

Students with a parent who has **at least one A-level** have a **56.8% chance** of saying they sat French.

This means:

Probability increases by **~6 percentage points** compared to the baseline

Confidence interval: **54.9% to 58.6%**

```r
# Predicted probabilities by s1pared
emmeans::emmeans(logmodel1, ~ s1pared, type = "response")
```

```
s1pared                         prob      SE  df asymp.LCL asymp.UCL
Neither parent with A-level    0.506 0.00623 Inf     0.494     0.519
At least one parent with A-level 0.568 0.00933 Inf     0.549     0.586
At least one parent with degree  0.635 0.00768 Inf     0.620     0.650
```

Getting the predicted probability

At least one parent has a degree → Probability = 0.635

Students with a parent who has a **degree** have a **63.5% chance** of saying they sat French.

This means:

Probability increases by **~13 percentage points** compared to the baseline

Confidence interval: **62.0% to 65.0%**

```
# Predicted probabilities by s1pared
emmeans::emmeans(logmodel1, ~ s1pared, type = "response")
```

```
s1pared                            prob      SE  df asymp.LCL asymp.UCL
Neither parent with A-level       0.506 0.00623 Inf     0.494     0.519
At least one parent with A-level 0.568 0.00933 Inf     0.549     0.586
At least one parent with degree  0.635 0.00768 Inf     0.620     0.650
```

Interpreting the output

The AIC loss function (2k-2*log(L)) tries handle the bias when fitting a model. When you fit a model if you increase the number of parameters you will improve the log likelihood but will run into the danger of over fitting. Use for the AIC comparing models.

```
Call:
glm(formula = satfren_num ~ s1pared, family = binomial(link = logit),
    data = youthdf, na.action = na.exclude)

Coefficients:
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```

Step 4: Assess the Classification Performance

We've predicted probabilities… but are they any good?

- A logistic regression gives us *numbers between 0 and 1*.
  But that does not tell us:
  - How often the model is **right**
  - How often it is **wrong**
- Whether it predicts **yes** better than **no**
- Or whether it is just doing something close to chance

To answer that, we need a way to evaluate the *quality* of the predictions.

That brings us to the **ROC curve, confusion matrix, sensitivity and specificity**

Step 4a: ROC Curve

Used to see how any predictive model can distinguish between the <u>true positives and negatives</u> by:

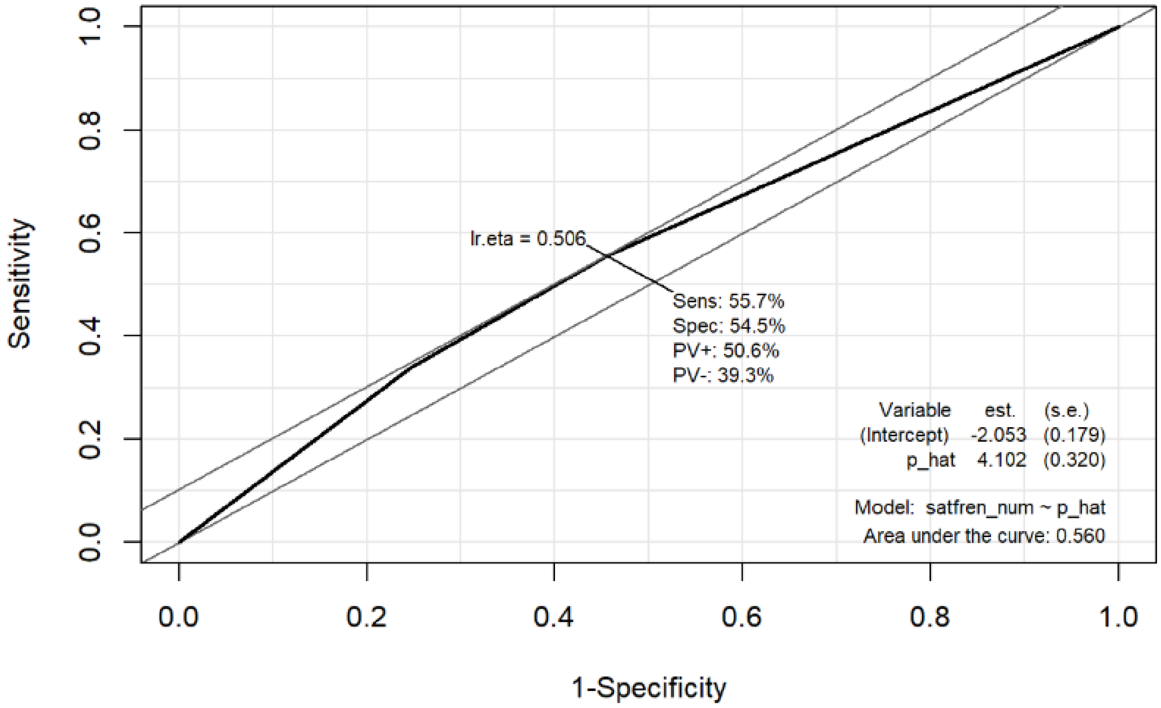
- plotting
  - ***Sensitivity***, the probability of predicting a real positive will be a positive,
- against
  - ***1-specificity***, the probability of predicting a real negative will be a positive.

The ROC curve plots out the sensitivity and specificity for every possible decision rule cutoff between 0 and 1 for a model Receiver Operating Characteristic.

Why do we need it:

- Visualize the classification of whether a student as “sat French” or “did not sit French”
- We pick a cut-off (e.g., 0.5) – but:
  - Changing the cut-off changes the model’s accuracy
  - Sensitivity and specificity depend on the cut-off
  - We need a way to evaluate the model across all possible cut-offs
  - The ROC curve does exactly that.

Interpreting the ROC



-Epi::ROC(form = satfren_num ~ p_hat, data = results1, plot = "ROC“)

A model that predicts at chance will have an ROC curve that looks like the lower diagonal line.

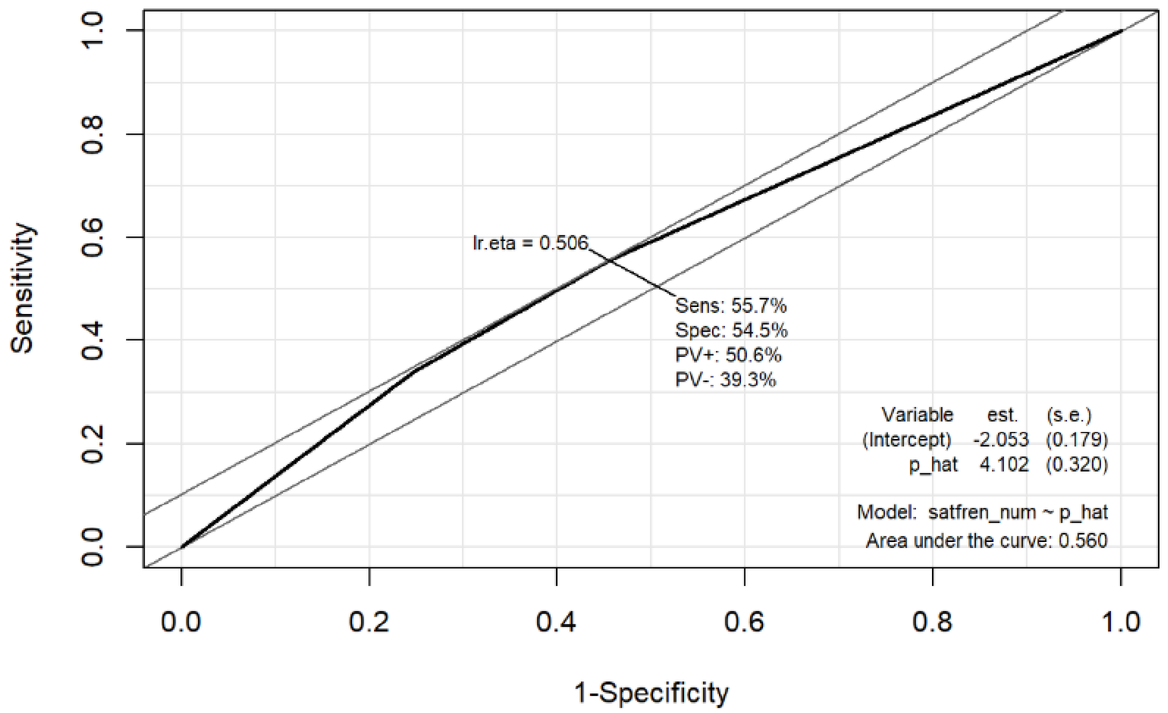
The further the curve is from the lower diagonal line, the better the model is at discriminating between positives and negatives in general.

Step 4a: AUC

To use the ROC curve to **quantify the performance of a classifier** use the **Area Under the Curve**.

AUC is literally just the percentage of this graph that is under this curve.

This AUC here is 0.56 (.5 is considered weak, 0.8 strong)



Step 4b: Confusion Matrix

A **confusion matrix** is a table that shows how well a classification model (like logistic regression) performed when predicting outcomes.

It compares:

- **What the model predicted**

  vs.

- **What actually happened**

This lets us see **where the model is getting things right and where it is making mistakes**

Confusion Matrix

		Test Indicator	
		No	Yes
Outcome	No	a True Negative	b False Positive
	Yes	c False Negative	d True Positive

Positive = student sat French (1)

Negative = student did not sit French (0)

True Positive (TP)

- The model predicted **Yes** and the student actually **did** sit French.
  ("Correct positive prediction")

False Negative (FN)

- The model predicted **No**, but the student actually **did** sit French.
  ("Missed positive")

False Positive (FP)

- The model predicted **Yes**, but the student actually **did not** sit French.
  ("False alarm")

True Negative (TN)

- The model predicted **No**, and the student really **did not** sit French.
  ("Correct negative prediction")

Sensitivity (true positive rate)

Tells us **how good the model is at correctly identifying the people who *did* sit French**.

It is the proportion of actual *yes* cases that the model correctly predicts as *yes*.

Formula:

$$\text{Sensitivity} = \frac{\mathit{True\ Positives}}{\mathit{True\ Positives} + \mathit{False\ Negatives}}$$

Test Indicator

Outcome		No	Yes
	No	a True Negative	b False Positive
	Yes	c False Negative	d True Positive

Specificity (true negative rate)

Tells us **how good the model is at correctly identifying the people who *did not* sit French**.

It is the proportion of actual *no* cases that the model correctly predicts as *no*.

Formula:

$$\text{Specificity} = \frac{\textit{True Negatives}}{\textit{True Negatives} + \textit{False Positives}}$$

		Test Indicator	
		No	Yes
Outcome	No	a True Negative	b False Positive
	Yes	c False Negative	d True Positive

Sensitivity and Specificity

A model that is so bad it's worthless would have a lot of b's and c's and possibly both.

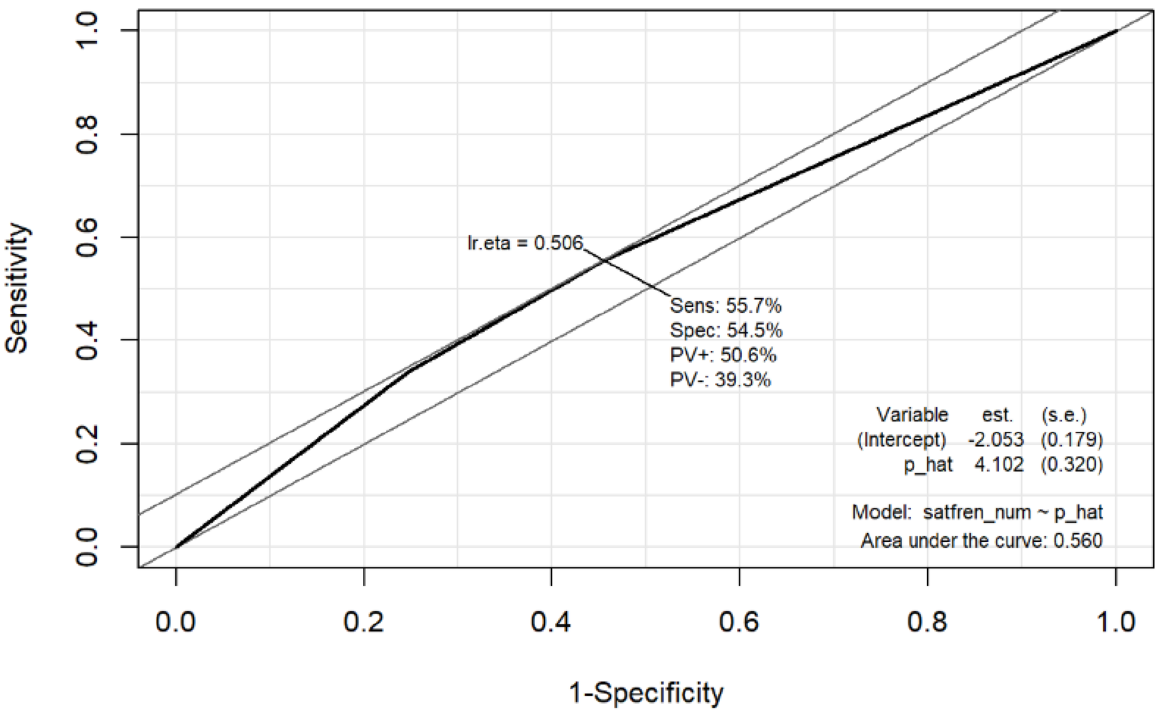
A simple way of measuring model accuracy is simply the proportion of individuals who were correctly classified–the proportions of True Positives and True Negatives.

We look at the accuracy of positives and negatives separately

		Test Indicator	
		No	Yes
Outcome	No	a True Negative	b False Positive
	Yes	c False Negative	d True Positive

Sensitivity

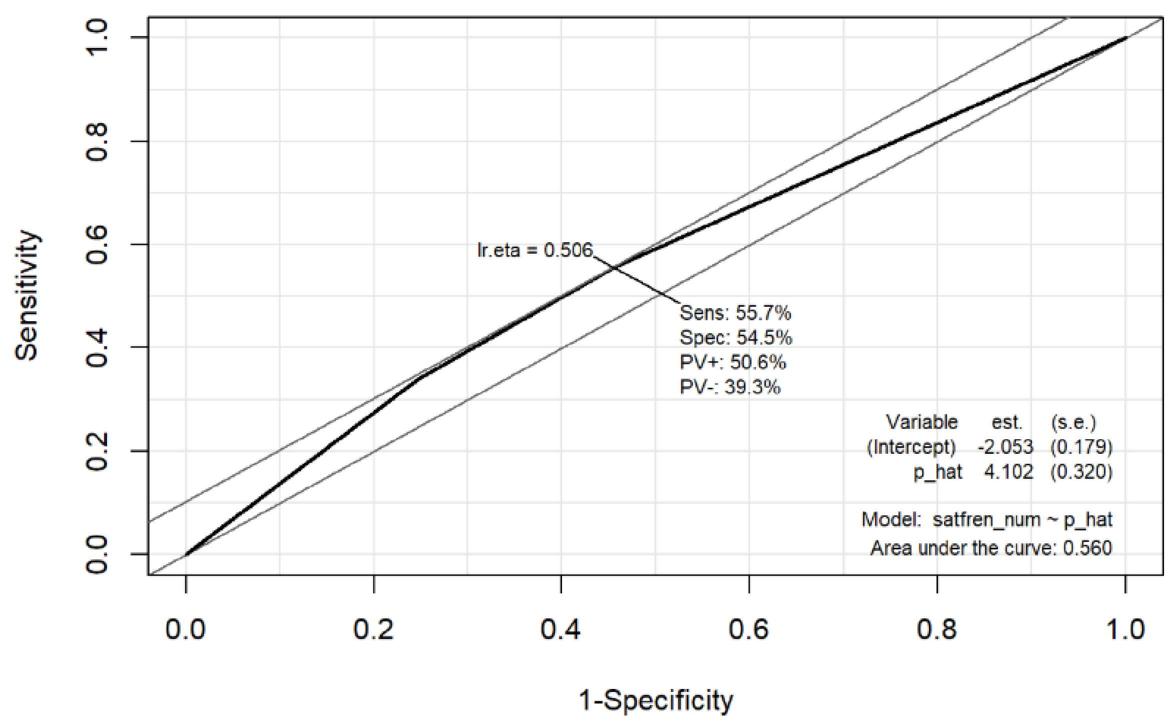
The ***sensitivity*** of the model is the percentage of the group *with the characteristic of interest* that has been accurately identified by the model



Sensitivity

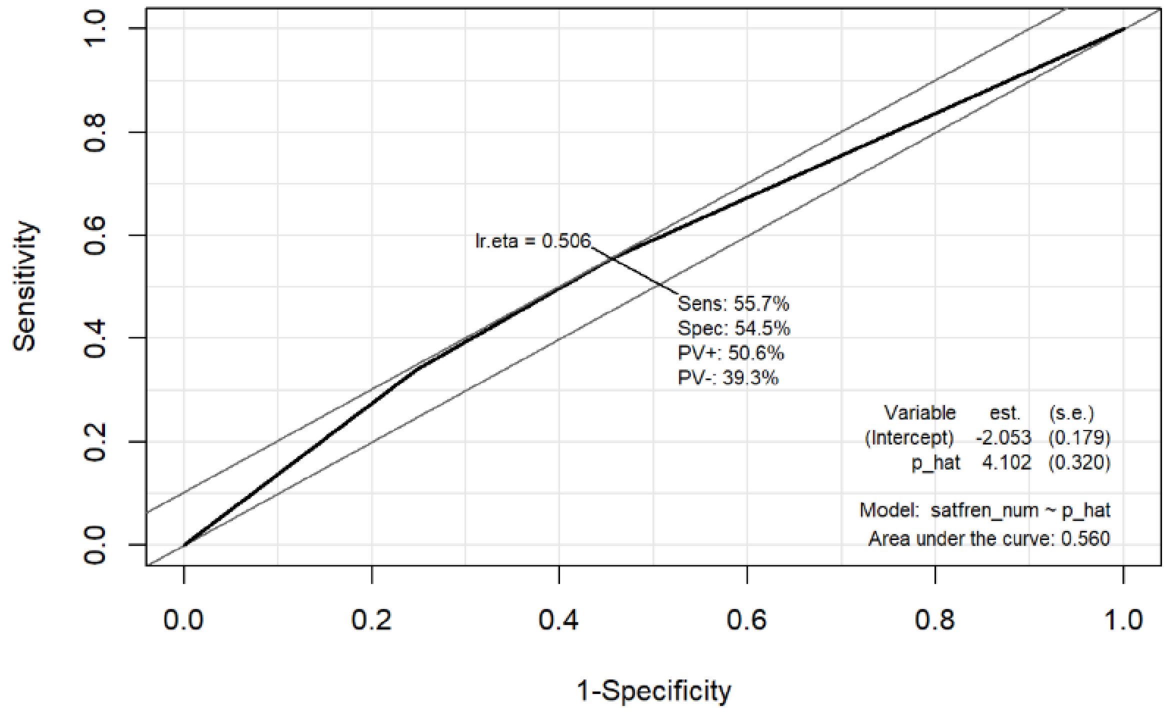
Those with "yes" for sitting French correctly predicted by the model (i.e., true positives).

In this case the model was able to identify 56% of the people who said they sat French.



Specificity

The ***specificity*** of the model is the percentage of the group *without the characteristic of interest*



Specificity

Those with "no" for sitting French and were also correctly predicted as not having the observed characteristic (i.e., true negatives).

In this case 54.5%.

Step 5a: Checking Assumptions- Linearity

```
#Check the assumption of linearity of independent variables and log odds using a Hosmer-Lemeshow
generalhoslem::logitgof(youthdf$satfren_num, fitted(logmodel1))
```

```
Warning in generalhoslem::logitgof(youthdf$satfren_num, fitted(logmodel1)): Not
possible to compute 10 rows. There might be too few observations.


    Hosmer and Lemeshow test (binary model)

data:  youthdf$satfren_num, fitted(logmodel1)
X-squared = 2.5495e-24, df = 0, p-value < 2.2e-16
```

Step 5b: Checking Assumptions- Collinearity

```
#We would check for collinearity here but as we only have one predictor it doesn't make sense -
```

Step 5c: Checking Assumptions- Outliers and Influential Cases

```
# 1. Cook's Distance
cook <- cooks.distance(logmodel1)

# View largest Cook's D values
head(sort(cook, decreasing = TRUE))
```

```
         50         119         121         122         156         200 
0.000155355 0.000155355 0.000155355 0.000155355 0.000155355 0.000155355 
```

```
# Flag influential cases (common threshold: D > 1)
which(cook > 1)
```

```
named integer(0)
```

Cook's Distance: Measures how much each individual observation influences the overall model

Large values indicate cases that may disproportionately affect the results.

No rows with Cooks distance > 1, no problems

Step 5c: Checking Assumptions- Outliers and Influential Cases

```r
# 2. Leverage (Hat values)
lev <- hatvalues(logmodel1)

# View highest leverage cases
head(sort(lev, decreasing = TRUE))
```

```
          10           13           17           19           50           66 
0.0003546099 0.0003546099 0.0003546099 0.0003546099 0.0003546099 0.0003546099 
```

```r
# Flag high leverage cases
# Rule of thumb: h_i > 2*(k+1)/n
k <- length(coef(logmodel1)) - 1  # number of predictors
n <- nobs(logmodel1)
lev_threshold <- 2*(k+1)/n
which(lev > lev_threshold)
```

```
named integer(0)
```

Leverage (Hat values): Identifies observations with unusual predictor values

High-leverage cases have the potential to pull the model toward themselves.

None found

Leverage Cases – What to Check/What to do?

Threshold: hi>2(k+1) /n

- Where: h_i =leverage for case I, k= number of predictors. n= sample size

Low leverage (< threshold)

- Case has typical predictor values
- Does not strongly affect the model
- No action needed

Moderate leverage (near threshold)

- Case has somewhat unusual predictor values
- Worth keeping an eye on
- Check Cook's Distance and DFBetas to confirm no problems

High leverage (> threshold, but < 0.2)

- Predictor pattern is unusual
- Case has potential to influence coefficients
- If Cook's Distance is low and DFBetas are small → not a problem

Very high leverage (> 0.2)

- Unusual case with strong potential to distort model
- Requires investigation
- Check raw data for entry errors, miscoding, or whether it represents a real subgroup

Step 5c: Checking Assumptions- Outliers and Influential Cases

```r
# 3. DFBetas (influence on individual coefficients)
dfb <- dfbeta(logmodel1)

# Identify cases with large DFBetas (|DFB| > 1 is concerning)
which(abs(dfb) > 1, arr.ind = TRUE)
```

```
     row col
```

DFBetas: Measure how much each observation influences individual regression coefficients;

Large DFBetas indicate that removing the case would substantially change a coefficient estimate.

None found

Step 5c: Checking Assumptions- Outliers and Influential Cases

```r
# 3. DFBetas (influence on individual coefficients)
dfb <- dfbeta(logmodel1)

# Identify cases with large DFBetas (|DFB| > 1 is concerning)
which(abs(dfb) > 1, arr.ind = TRUE)
```

```
     row col
```

Standardised (Pearson) residuals:

Show how poorly each observation is predicted by the model

Large residuals (e.g., > |2|) indicate unusual or poorly fitted cases.

None found

Step 5c: Checking Assumptions- Outliers and Influential Cases

```r
# 5. Deviance residuals
dev_resid <- residuals(logmodel1, type = "deviance")

# Flag extreme deviance residuals
which(abs(dev_resid) > 2)
```

```
named integer(0)
```

Deviance residuals: Reflect how much each observation contributes to the model’s deviance

Large values highlight observations the model struggles to accurately classify.

None found.

Interpreting the output

The AIC loss function (2k-2*log(L)) tries handle the bias when fitting a model. When you fit a model if you increase the number of parameters you will improve the log likelihood but will run into the danger of over fitting. Use for the AIC comparing models.

```
Call:
glm(formula = satfren_num ~ s1pared, family = binomial(link = logit),
    data = youthdf, na.action = na.exclude)

Coefficients:
                                        Estimate Std. Error z value Pr(>|z|)
(Intercept)                              0.02544    0.02491   1.021    0.307
s1paredAt least one parent with A-level  0.24716    0.04545   5.439 5.37e-08
s1paredAt least one parent with degree   0.52779    0.04144  12.736  < 2e-16

(Intercept)
s1paredAt least one parent with A-level ***
s1paredAt least one parent with degree  ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 18124  on 13200  degrees of freedom
Residual deviance: 17958  on 13198  degrees of freedom
AIC: 17964
```

Reporting

See the report in MATH9102-W11-LogReg.html

Time to try it out yourself

Download the files from Brightspace.

Open MATH9102-W11-LogReg-Exercise.qmd

TASK 1:

Using the Sleep dataset build a logistic regression model to predict the probability of being in the category of having trouble sleeping (variable trubslep 1 = yes; 2 = no) with the predictors of smoking (variable smoke 1 = yes; 2 = no)

Review the output and interpret.

Extending the youthcohort model to include Secondary School Type as a predictor

Categorical variable

- pcschtype
- Values
  - Comprehensive 16
  - Comprehensive 18
  - CTC/Other maintained
  - Grammar
  - Independent
  - Modern

Model Including s1pared

```
logmodel2 <- glm(
satfren_num ~ s1pared +
pcschtype,  data =
youthdf,  family =
stats::binomial(link =
"logit"),  na.action =
stats::na.exclude)
```

	Dependent variable:
	satfren_num
s1paredAt least one parent with A-level	0.204***
	(0.046)
s1paredAt least one parent with degree	0.381***
	(0.043)
pcschtypeComprehensive 18	0.078
	(0.040)
pcschtypeCTC/Other maintained	-0.391*
	(0.191)
pcschtypeGrammar	0.302***
	(0.084)
pcschtypeIndependent	1.037***
	(0.078)
pcschtypeModern	0.321**
	(0.110)
Constant	-0.067
	(0.035)
Observations	13,201
Log Likelihood	-8,868.015
Akaike Inf. Crit.	17,752.030
Note:	*p<0.05; **p<0.01; ***p<0.001

```
Likelihood ratio test

Model 1: satfren_num ~ s1pared + pcschtype
Model 2: satfren_num ~ 1
  #Df LogLik Df  Chisq Pr(>Chisq)
1   8  -8868
2   1  -9062 -7 387.88  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Pseudo Rsquared
DescTools::PseudoR2(logmodel2, which="CoxSnell")
```

```
  CoxSnell
0.02895493
```

```
DescTools::PseudoR2(logmodel2, which="Nagelkerke")
```

```
Nagelkerke
0.03878063
```

```
performance::r2(logmodel1) # Will return Tujur's Pseudo R2
```

```
# R2 for Logistic Regression
  Tjur's R2: 0.013
```

How useful is the model?

MODEL IS SIGNIFICANT CHI SQ=387.88 P<0.01, EXPLAINING BETWEEN 1.3 AND 3.9% OF THE VARIATION

Interpreting the coefficients

```
Call:
glm(formula = satfren_num ~ s1pared + pcschtype, family = stats::binomial(link = "logit"),
    data = youthdf, na.action = stats::na.exclude)

Coefficients:
                                      Estimate Std. Error z value Pr(>|z|)
(Intercept)                           -0.06652    0.03463  -1.921 0.054769
s1paredAt least one parent with A-level  0.20421    0.04590   4.449 8.64e-06
s1paredAt least one parent with degree   0.38101    0.04311   8.838  < 2e-16
pcschtypeComprehensive 18              0.07760    0.03989   1.945 0.051756
pcschtypeCTC/Other maintained         -0.39117    0.19114  -2.046 0.040709
pcschtypeGrammar                       0.30223    0.08410   3.594 0.000326
pcschtypeIndependent                   1.03692    0.07775  13.336  < 2e-16
pcschtypeModern                        0.32112    0.11018   2.914 0.003563

(Intercept)                             .
s1paredAt least one parent with A-level ***
s1paredAt least one parent with degree  ***
pcschtypeComprehensive 18               .
pcschtypeCTC/Other maintained           *
pcschtypeGrammar                        ***
pcschtypeIndependent                    ***
pcschtypeModern                         **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Odds Ratios

```
                                Parameter OddsRatio
1                             (Intercept) 0.9356409
2 s1paredAt least one parent with A-level 1.2265617
3  s1paredAt least one parent with degree 1.4637632
4              pcschtypeComprehensive 18 1.0806920
5          pcschtypeCTC/Other maintained 0.6762653
6                        pcschtypeGrammar 1.3528704
7                    pcschtypeIndependent 2.8205052
8                         pcschtypeModern 1.3786730
```

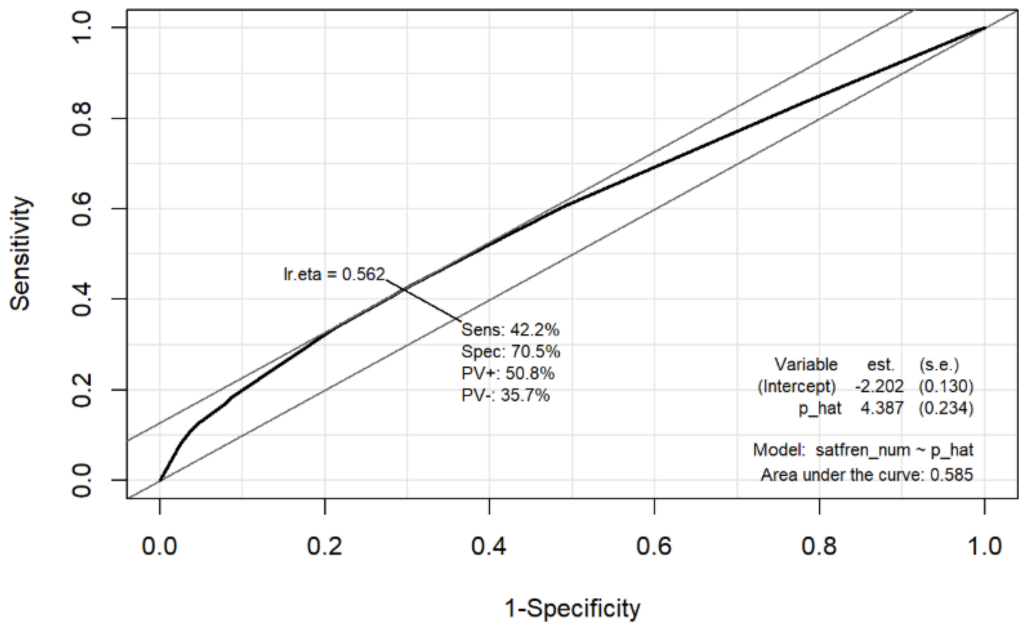
Predicted Probabilities

```
pred_grid <- emmeans::emmeans(
  logmodel2,
  ~ s1pared * pcschtype,   # <-- interaction in predictions, even if not in model
  type = "response"
)

summary(pred_grid)
```

```
s1pared                          pcschtype                prob      SE  df
Neither parent with A-level      Comprehensive 16        0.483 0.00865 Inf
At least one parent with A-level Comprehensive 16        0.534 0.01160 Inf
At least one parent with degree  Comprehensive 16        0.578 0.01100 Inf
Neither parent with A-level      Comprehensive 18        0.503 0.00762 Inf
At least one parent with A-level Comprehensive 18        0.554 0.01030 Inf
At least one parent with degree  Comprehensive 18        0.597 0.00926 Inf
Neither parent with A-level      CTC/Other maintained 0.388 0.04500 Inf
At least one parent with A-level CTC/Other maintained 0.437 0.04710 Inf
At least one parent with degree  CTC/Other maintained 0.481 0.04770 Inf
Neither parent with A-level      Grammar                 0.559 0.02010 Inf
At least one parent with A-level Grammar                 0.608 0.02010 Inf
At least one parent with degree  Grammar                 0.649 0.01820 Inf
Neither parent with A-level      Independent             0.725 0.01500 Inf
At least one parent with A-level Independent             0.764 0.01420 Inf
At least one parent with degree  Independent             0.794 0.01170 Inf
Neither parent with A-level      Modern                  0.563 0.02630 Inf
At least one parent with A-level Modern                  0.613 0.02630 Inf
At least one parent with degree  Modern                  0.654 0.02510 Inf
```

ROC Curve



Sensitivity
1-Specificity
0.0
0.2
0.4
0.6
0.8
1.0
lr.eta = 0.562
Sens: 42.2%
Spec: 70.5%
PV+: 50.8%
PV-: 35.7%
Variable est. (s.e.)
(Intercept) -2.202 (0.130)
p_hat 4.387 (0.234)
Model: satfren_num ~ p_hat
Area under the curve: 0.585

Confusion matrix

```
             Truth
Prediction    0    1
         0 1300 1229
         1 4538 6134
```

Checking Assumptions- Linearity

```
#Check the assumption of linearity of independent variables and log odds using a Hosmer-Lemeshow

generalhoslem::logitgof(youthdf$satfren_num, fitted(logmodel2))
```

```
Warning in generalhoslem::logitgof(youthdf$satfren_num, fitted(logmodel2)): Not
possible to compute 10 rows. There might be too few observations.


    Hosmer and Lemeshow test (binary model)

data:  youthdf$satfren_num, fitted(logmodel2)
X-squared = 3.0885, df = 6, p-value = 0.7977
```

Checking Assumptions - Collinearity

Tolerance < 0.1 → Serious multicollinearity

Tolerance < 0.2 → Potential multicollinearity problem

VIF > 10 → Serious multicollinearity (widely used rule of thumb)

VIF > 5 → Concerning / needs investigation

VIF between 1 and 5 → Acceptable

```
#Collinearity
vifmodel<-car::vif(logmodel2)
# You can ignore the warning messages
# GVIF^(1/(2*Df)) is the value of interest

vifmodel
```

```
              GVIF Df GVIF^(1/(2*Df))
s1pared   1.060265  2        1.014737
pcschtype 1.060265  5        1.005869
```

```
#Tolerance
1/vifmodel
```

```
               GVIF  Df GVIF^(1/(2*Df))
s1pared   0.9431604 0.5       0.9854768
pcschtype 0.9431604 0.2       0.9941652
```

Outliers and Influential Cases

```r
# 1. Cook's Distance
cook <- cooks.distance(logmodel2)

# View largest Cook's D values
head(sort(cook, decreasing = TRUE))
```

```
       1286        1562        1967        2384        2682        3569 
0.001711676 0.001711676 0.001711676 0.001711676 0.001711676 0.001711676 
```

```r
# Flag influential cases (common threshold: D > 1)
which(cook > 1)
```

```
named integer(0)
```

Outliers and Influential Cases

This is a problem!

We need to check the other diagnostics

```
# 2. Leverage (Hat values)
lev <- hatvalues(logmodel2)

# View highest leverage cases
head(sort(lev, decreasing = TRUE))
```

```
       1424        3370        3371        3372        3566        3567 
0.009107665 0.009107665 0.009107665 0.009107665 0.009107665 0.009107665 
```

```
#Count how many exceed threshold
sum(lev > lev_threshold)
```

```
[1] 4123
```

```
#Count how many > .2
sum(lev > 0.2)
```

```
[1] 0
```

Outliers and Influential Cases

```r
# 3. DFBetas (influence on individual coefficients)
dfb <- dfbeta(logmodel2)

# Identify cases with large DFBetas (|DFB| > 1 is concerning)
which(abs(dfb) > 1, arr.ind = TRUE)
```

```
     row col
```

```r
# 4. Standardised (Pearson) residuals
pearson_resid <- rstandard(logmodel2, type = "pearson")

# Flag large residuals (|residual| > 2 or > 3)
which(abs(pearson_resid) > 2)
```

```
named integer(0)
```

Outliers and Influential Cases

```r
# 5. Deviance residuals
dev_resid <- residuals(logmodel2, type = "deviance")

# Flag extreme deviance residuals
which(abs(dev_resid) > 2)
```

```
named integer(0)
```

Outliers and Influential Cases

Because there are no other issues – we can assume the leverage cases are not a problem

High leverage (> threshold, but < 0.2)

- Predictor pattern is unusual
- Case has potential to influence coefficients
- If Cook's Distance is low and DFBetas are small → not a proble

```
# 2. Leverage (Hat values)
lev <- hatvalues(logmodel2)

# View highest leverage cases
head(sort(lev, decreasing = TRUE))
```

```
       1424        3370        3371        3372        3566        3567 
0.009107665 0.009107665 0.009107665 0.009107665 0.009107665 0.009107665 
```

```
#Count how many exceed threshold
sum(lev > lev_threshold)
```

```
[1] 4123
```

```
#Count how many > .2
sum(lev > 0.2)
```

```
[1] 0
```

Interpreting the output

The AIC loss function (2k-2*log(L)) tries handle the bias when fitting a model. When you fit a model if you increase the number of parameters you will improve the log likelihood but will run into the danger of over fitting.

Use for the AIC comparing models.

Lower AIC values represent preferable models, so this reduction provides strong evidence that adding school type improves model quality in comparison to model1

```
Call:
glm(formula = satfren_num ~ s1pared + pcschtype, family = stats::binomial(link =
    data = youthdf, na.action = stats::na.exclude)

Coefficients:
                                      Estimate Std. Error z value Pr(>|z|)
(Intercept)                           -0.06652    0.03463  -1.921 0.054769
s1paredAt least one parent with A-level  0.20421    0.04590   4.449 8.64e-06
s1paredAt least one parent with degree   0.38101    0.04311   8.838  < 2e-16
pcschtypeComprehensive 18              0.07760    0.03989   1.945 0.051756
pcschtypeCTC/Other maintained         -0.39117    0.19114  -2.046 0.040709
pcschtypeGrammar                       0.30223    0.08410   3.594 0.000326
pcschtypeIndependent                   1.03692    0.07775  13.336  < 2e-16
pcschtypeModern                        0.32112    0.11018   2.914 0.003563

(Intercept)                           .
s1paredAt least one parent with A-level ***
s1paredAt least one parent with degree  ***
pcschtypeComprehensive 18             .
pcschtypeCTC/Other maintained         *
pcschtypeGrammar                      ***
pcschtypeIndependent                  ***
pcschtypeModern                       **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 18124  on 13200  degrees of freedom
Residual deviance: 17736  on 13193  degrees of freedom
AIC: 17752
```

Comparing Models

Model Fit (Overall Performance)

- Likelihood Ratio (LR) Test
  - Does the fitted model significantly improve fit over the previous model?
  - Compare χ^2 values between models.
- Akaike Information Criterion (AIC)
  - Lower values indicate better balance of fit and complexity.

Pseudo R^2 Measures

- Cox–Snell R^2
- Nagelkerke R^2
- Tjur's R^2
- Higher values indicate better explanatory power (but not variance explained like linear R^2).

Comparing Models

Coefficient Significance and Effect Sizes

- Are predictors statistically significant? Does this change across models?
- Do odds ratios increase or decrease across models?
- Do effect sizes change (e.g., become smaller after adding new predictors)?

Predicted Probabilities

- Are predicted probabilities more distinct across categories?
- Does a more complex model capture more meaningful differences between groups?

Interpretation and Theoretical Meaning

- Do added predictors make sense?
- Does the model with more predictors tell a clearer or more informative story?

Comparing Models

Classification Performance

- ROC curve
- Area Under the Curve (AUC)
- Sensitivity (true positive rate)
- Specificity (true negative rate)
- Confusion matrix patterns
- Does one model classify cases more accurately?

Model Parsimony (Simplicity vs Usefulness)

- Does adding predictors meaningfully improve fit and interpretation?
- Is the added complexity justified?

Comparing Models

Assumptions and Diagnostics

- Hosmer–Lemeshow goodness-of-fit
- Multicollinearity (VIF/tolerance)
- Outliers or influential cases
- Does one model meet assumptions better than another?

Time to try for yourself

MATH 9102-W11-LogReg-Exercise.qmd

TASK 2 : Extend your model including total number of caffeine drinks as a predictor.

What does logistic regression try to do?

Model the probability of an event occurring depending on the values of the independent variables

Estimate the probability that an event occurs vs the probability that the event does not occur

Predict the effect of a series of variables on a binary response variable

Classify observations by estimating the probability that an observation is in a particular category or not.

Assumptions

Requires the dependent variable to be binary/nominal with multiple categories

- ordinal logistic regression requires the dependent variable to be ordinal.

Requires the observations to be independent of each other.

- Observations should not come from repeated measurements or matched data.

Requires there to be little or no multicollinearity among the independent variables.

- Independent variables should not be too highly correlated with each other.
- Check Post-hoc

Requires linear relationship between the log odds and the predictors.

- Check Post-hoc

Assumptions

Typically requires a large sample size.

- A general guideline is that you need at minimum of 10 cases with the least frequent outcome for each independent variable in your model.
  - For example, if you have 5 independent variables and the expected probability of your least frequent outcome is .10, then you would need a minimum sample size of 500 (10*5 / .10).

Multinomial logistic regression

Logistic regression to predict membership of more than two categories.

It (basically) works in the same way as binary logistic regression.

The analysis breaks the outcome variable down into a series of comparisons between two categories.

- E.g., if you have three outcome categories (A, B and C), then the analysis will consist of two comparisons that you choose:
  - Compare everything against your first category (e.g. A vs. B and A vs. C),
  - Or your last category (e.g. A vs. C and B vs. C),
  - Or a custom category (e.g. B vs. A and B vs. C).

The important parts of the analysis and output are much the same as we have just seen for binary logistic regression

How to report Logistic Regression – Example

A multinomial logistic regression analysis was conducted with a student's level study as the outcome variable (undergraduate, taught postgraduate, research postgraduate) with student well-being, mature student status and the type of educational institute being attended as predictors.

The data met the assumption for independent observations. Examination for multicollinearity showed that the tolerance and variance influence factor measures were within acceptable levels (tolerance >0.4, VIF <2.5) as outlined in Tarling (2008). The Hosmer Lemeshow goodness of fit statistic did not indicate any issues with the assumption of linearity between the independent variables and the log odds of the model ($\chi^2(n=8)=$ =11.09, p =0.2).

*we are not doing this.